

Digital Audio, Lossless & Lossy Compression, and Audio Coding

From Sound Waves to MP3

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Outline

- 1 Basics of Digital Audio
- 2 MIDI
- 3 Lossless Compression Algorithms
- 4 Distortion Measures & Rate-Distortion Theory
- 5 Quantization
- 6 Transform Coding & Wavelets
- 7 Audio Compression Techniques
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- 10 Summary

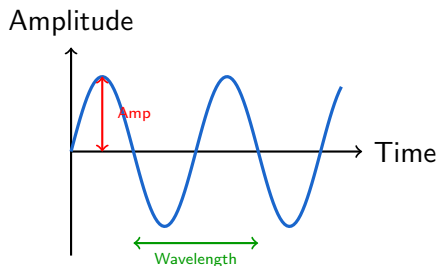
What is Sound?

Sound is a pressure wave travelling through air.

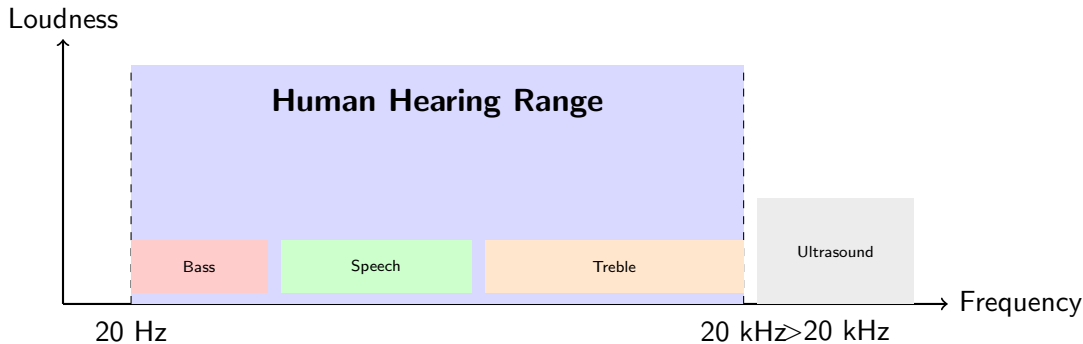
- Vibrating object pushes air molecules
- Creates alternating high/low pressure regions
- Our eardrums vibrate in response
- Brain interprets vibrations as sound

Key properties:

- **Frequency** (Hz) — pitch (low/high)
- **Amplitude** — loudness (quiet/loud)
- **Timbre** — tone colour (guitar vs piano)

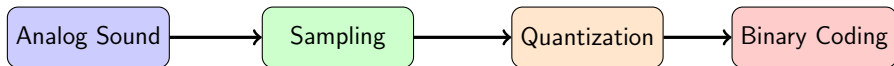


Human Hearing Range



- Speech occupies roughly 300 Hz – 3400 Hz (telephone band)
- Music spans the full 20 Hz – 20 kHz range
- Sensitivity peaks around 2–5 kHz (where speech consonants live)

Digitization of Sound: Three Steps



1. Sampling

Measure the wave at regular intervals.

How often?

2. Quantization

Round each measurement to the nearest allowed level.

How precise?

3. Binary Coding

Represent each quantized value as a binary number.

How many bits?

Sampling and the Nyquist Theorem

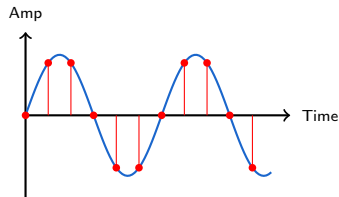
Nyquist Theorem

To perfectly reconstruct a signal of bandwidth B , sample at rate:

$$f_s \geq 2B$$

Common sampling rates:

Use	Rate	Bandwidth
Telephone	8 kHz	4 kHz
AM Radio	22.05 kHz	11 kHz
CD Audio	44.1 kHz	22 kHz
DVD/Studio	96 kHz	48 kHz



Red dots = samples taken at regular intervals. More samples = better reconstruction.

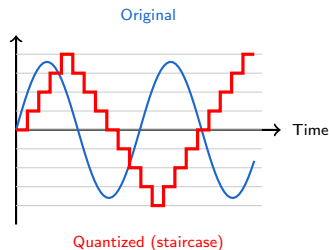
Quantization and Bit Depth

Quantization maps continuous amplitude to discrete levels.

Bit depth determines the number of levels:

Bits	Levels	Use
8	256	Telephone
16	65,536	CD Audio
24	16.7M	Studio

Dynamic range: ≈ 6 dB per bit.
16-bit $\rightarrow$ 96 dB (excellent for music).



Audio Data Rates

Uncompressed audio data rate:

$$\text{Rate} = f_s \times \text{bits} \times \text{channels}$$

Format	Rate	Bits	Ch	Bitrate
Telephone	8 kHz	8	1	64 kbps
FM Radio	22 kHz	16	1	352 kbps
CD Audio	44.1 kHz	16	2	1,411 kbps
DVD Audio	96 kHz	24	6	13,824 kbps

The Problem

One minute of CD audio = $\frac{1411 \times 60}{8} \approx$ **10.6 MB**.

A 50-minute album = **530 MB**. That's why we need compression!

Musical Instrument Digital Interface (MIDI)

What is MIDI?

MIDI is not audio! It is a **set of instructions** that tell a synthesizer which notes to play, how loud, and for how long.

MIDI message example:

Field	Value
Note On	Channel 1
Pitch	C4 (middle C)
Velocity	100 (loud)
Duration	500 ms
Note Off	Channel 1

Advantages:

- Tiny file size (a few KB)
- Easy to edit (change tempo, key)
- 128 standard instruments
- 16 channels simultaneously

Limitation:

Sound quality depends on the synthesizer, not the file.

MIDI vs Digital Audio

Property	MIDI	Digital Audio
Stores	Instructions	Actual sound
File size (1 min)	~10 KB	~10 MB
Edit pitch/tempo	Easy	Difficult
Voice/speech	Cannot represent	Yes
Sound quality	Depends on synth	Fixed at recording
Standard	General MIDI	WAV, AIFF

Analogy

MIDI is like **sheet music** — instructions for a musician.

Digital audio is like a **recording** — the actual performance captured.

Why Lossless Compression?

Lossless = perfect reconstruction.

Not a single bit is lost.

Used when:

- Medical data
- Legal documents
- Software distribution
- Archival audio (FLAC)
- Text files

Typical compression ratios:

Data type	Ratio
English text	2:1 – 3:1
Source code	2:1 – 4:1
Audio (FLAC)	1.5:1 – 3:1
Images (PNG)	1.5:1 – 5:1
Random data	$\approx$ 1:1

Random data cannot be compressed — this is a fundamental limit!

Information Theory: Entropy

Shannon Entropy (1948)

The average information content of a source:

$$H(X) = - \sum_{i=1}^n p_i \log_2 p_i \quad (\text{bits/symbol})$$

Entropy sets the absolute limit on lossless compression.

No lossless code can achieve an average length below $H(X)$.

Quick example: Source with $P(A) = 0.5$, $P(B) = 0.25$, $P(C) = 0.125$, $P(D) = 0.125$

$$H = -(0.5 \cdot (-1) + 0.25 \cdot (-2) + 0.125 \cdot (-3) + 0.125 \cdot (-3)) = 1.75 \text{ bits/symbol}$$

We cannot compress below 1.75 bits per symbol on average.

Lossless Coding Techniques (Recap)

Method	Approach	Optimal?	Used in
Shannon-Fano	Top-down divide	No	Historical
Huffman	Bottom-up merge	Among prefix	JPEG, ZIP
Arithmetic	Interval narrowing	Yes	JPEG2000, H.264
LZW	Dictionary	Adaptive	GIF, Unix compress

Key ideas we covered in earlier lectures:

- Frequent symbols $\rightarrow$ short codes (variable-length coding)
- Prefix-free property ensures unambiguous decoding
- Arithmetic coding achieves fractional bits per symbol
- Dictionary methods exploit repeated patterns

Lossy Compression: Allowing Some Distortion

Key Idea

If we accept **some loss of quality**, we can compress far beyond the entropy limit.

But how do we measure “quality loss”?

Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2$$

Average squared difference between original x and reconstructed $\hat{x}$.

Signal-to-Noise Ratio (SNR):

$$\text{SNR} = 10 \log_{10} \frac{\sigma_x^2}{\text{MSE}} \text{ dB}$$

Higher SNR = better quality.

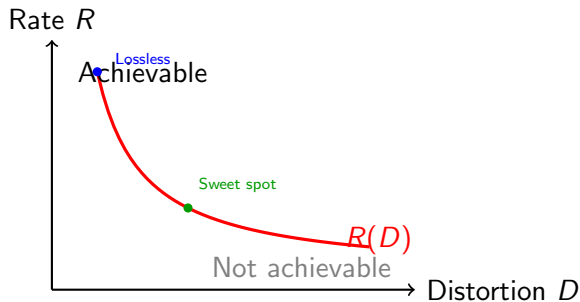
Typical targets:

- Speech: 20–30 dB
- Music: 60–90 dB

Rate-Distortion Theory

The Fundamental Tradeoff

Given a source and a distortion measure, $R(D)$ gives the minimum bit rate to represent the source with distortion $\leq D$.



More bits = less distortion. The curve tells us the best possible tradeoff.

Quantization: The Core of Lossy Compression

Definition

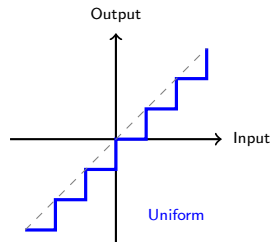
Quantization maps a large (possibly infinite) set of input values to a smaller, finite set of output values. This is inherently lossy.

Uniform Quantization:

- Equal-width steps
- Simple to implement

Non-uniform Quantization:

- Smaller steps where signal is common
- Better SNR for same bit rate



μ -Law and A-Law Companding

Used in telephone systems worldwide:

μ -Law (North America, Japan):

$$F(x) = \text{sgn}(x) \frac{\ln(1 + \mu|x|)}{\ln(1 + \mu)}$$

Typically $\mu = 255$.

A-Law (Europe, rest of world):

$$F(x) = \text{sgn}(x) \times \begin{cases} \frac{A|x|}{1 + \ln A} & |x| < \frac{1}{A} \\ \frac{1 + \ln(A|x|)}{1 + \ln A} & \text{else} \end{cases}$$

Typically $A = 87.6$.

Why companding?

- Speech is mostly quiet
- Fine steps for quiet sounds
- Coarse steps for loud sounds
- 8-bit companded $\approx$ 12-bit uniform quality

Result

Telephone quality at only 64 kbps (8 kHz $\times$ 8 bits).

Vector Quantization

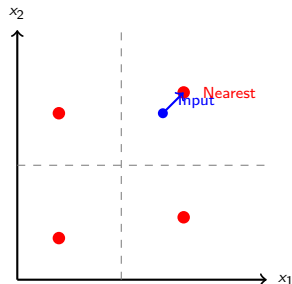
Quantize groups of samples together, not one at a time.

Scalar vs Vector:

- Scalar: quantize each sample independently
- Vector: quantize blocks of k samples as a single unit
- Vector is always $\geq$ as efficient as scalar

Codebook:

- Pre-computed set of representative vectors
- Encoder finds closest codebook entry
- Sends only the index

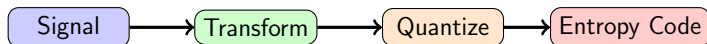


2D example: four codebook vectors partition the space into four regions.

Transform Coding

Core Idea

Transform the signal into a different domain where energy is concentrated in a few coefficients. Quantize and code only the important ones.



Common transforms:

- **DCT** (Discrete Cosine Transform) — used in JPEG, MP3
- **DFT** (Discrete Fourier Transform) — general frequency analysis
- **DWT** (Discrete Wavelet Transform) — used in JPEG 2000

Why Transforms Help Compression

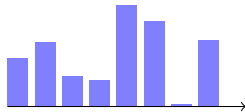
Before transform:

- Energy spread across all samples
- Every sample seems important
- Hard to decide what to discard

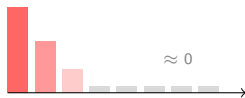
After transform:

- Energy packed into few coefficients
- Most coefficients are near zero
- Safe to discard the small ones

Before (time domain)



After (frequency domain)



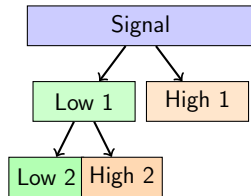
Wavelet-Based Coding

Wavelets vs DCT:

- DCT uses fixed-size blocks (e.g. 8×8)
- Wavelets use multi-resolution analysis
- Better at capturing transients (sharp attacks in audio)
- No blocking artifacts

How it works:

- 1 Split signal into low-frequency (approximation) and high-frequency (detail) parts
- 2 Repeat on the low-frequency part
- 3 Creates a multi-level decomposition

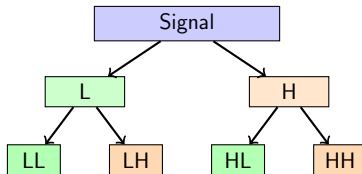


Each level halves the frequency resolution and doubles the time resolution.

Wavelet Packets

Extension of Standard Wavelets

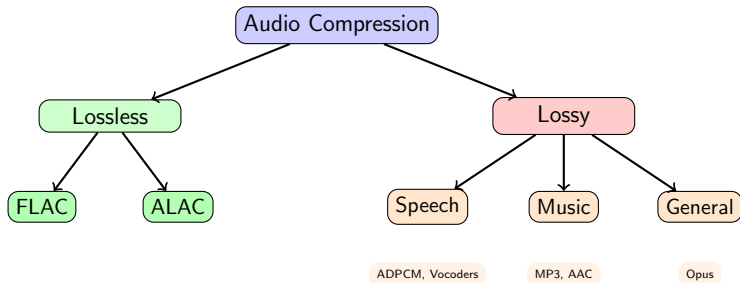
Standard wavelets only decompose the low-frequency part at each level. Wavelet packets decompose **both** low and high frequency parts, giving a richer set of bases.



Advantage: The best decomposition tree can be chosen adaptively for each signal, giving better energy compaction than a fixed wavelet decomposition.

Used in some advanced audio coders and JPEG 2000.

Audio Compression: The Landscape



ADPCM in Speech Coding

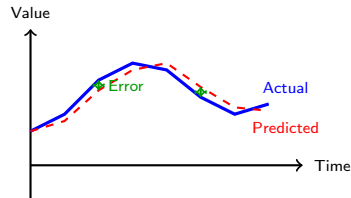
Adaptive Differential Pulse Code Modulation

Instead of coding the signal itself, code the **difference** between the actual sample and a **prediction** of it.

How ADPCM works:

- 1 Predict next sample from previous ones
- 2 Compute prediction error (residual)
- 3 Quantize the error (much smaller!)
- 4 Adapt step size based on signal

Adjacent speech samples are highly correlated, so the difference is small and needs fewer bits.

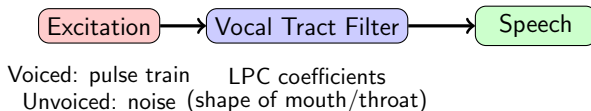


G.726 achieves toll-quality speech at **32 kbps** (half of PCM's 64 kbps).

Vocoders: Modelling the Voice

Vocoder = Voice Coder

Instead of coding the waveform, model the **speech production system** (vocal tract) and transmit only the model parameters.



Extremely low bitrates: 2.4 – 9.6 kbps (vs 64 kbps for PCM).

Tradeoff: Sounds robotic at very low rates. Good for military, satellite.

Waveform vs Model-Based vs Hybrid

	Waveform	Vocoder	Hybrid
Approach	Code signal	Code model	Both
Example	PCM, ADPCM	LPC vocoder	CELP
Bitrate	16–64 kbps	2.4–4.8 kbps	4.8–16 kbps
Quality	Natural	Robotic	Good
Non-speech	Good	Poor	Fair

CELP (Code-Excited Linear Prediction)

The dominant speech coding paradigm today. Used in GSM, 3G, 4G, VoLTE.
Combines LPC modelling with a codebook of excitation patterns.
Achieves near-toll quality at 8–13 kbps.

Psychoacoustics: What You Can't Hear

Key Insight

The human ear has limitations. If we know what you **cannot hear**, we can throw it away and save bits!

Two main phenomena:

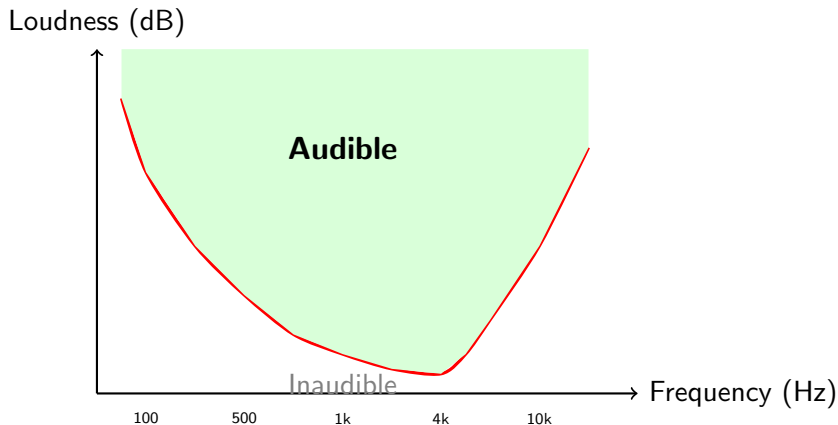
1. Absolute Threshold of Hearing

- Minimum loudness to hear a tone
- Varies with frequency
- Most sensitive at 2–5 kHz
- Anything below the threshold is inaudible

2. Masking

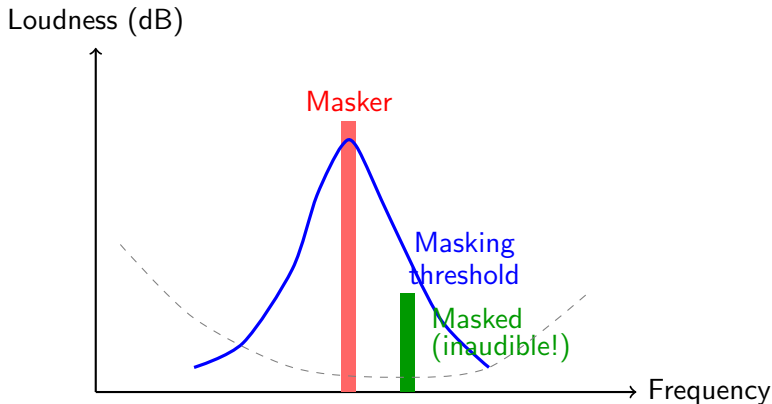
- A loud sound hides nearby quiet sounds
- Frequency masking (simultaneous)
- Temporal masking (before/after)
- The hidden sounds can be removed!

Absolute Threshold of Hearing



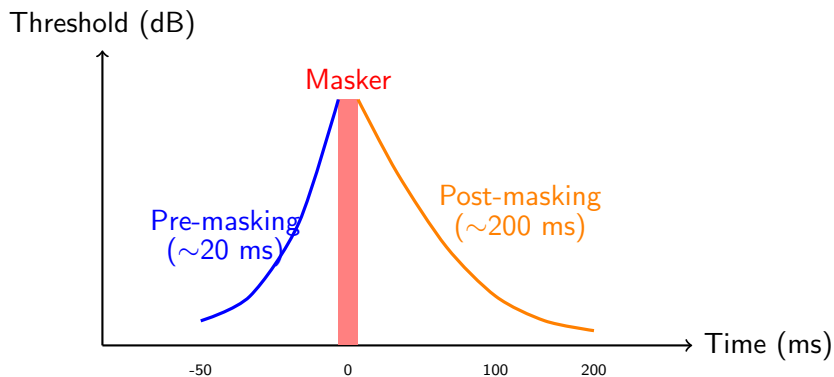
Sounds below the red curve are **inaudible** — no need to encode them!

Frequency Masking



A loud tone at one frequency raises the hearing threshold for nearby frequencies. Quiet tones near a loud one become **inaudible** and can be discarded.

Temporal Masking



A loud sound masks quiet sounds that occur **just before** (pre-masking, ~ 20 ms) and **after** it (post-masking, ~ 200 ms).

MPEG Audio: Three Layers

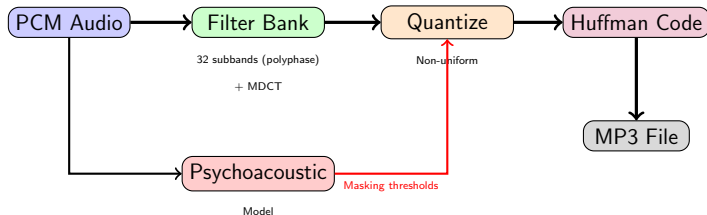
	Layer I	Layer II	Layer III (MP3)
Complexity	Low	Medium	High
Quality at 128 kbps	Fair	Good	Excellent
Typical bitrate	384 kbps	192 kbps	128 kbps
Compression ratio	4:1	8:1	12:1
Use	DCC tape	DAB radio	Internet, players

MP3 = MPEG-1 Audio Layer III

The most popular audio format in history. Achieves near-CD quality at 128–192 kbps (vs 1411 kbps for CD).

Key idea: Combine transform coding with psychoacoustic modelling.

MP3 Encoder Pipeline



Key steps:

- 1 **Filter bank:** Split audio into 576 frequency bands ($32 \text{ subbands} \times 18 \text{ MDCT}$)
- 2 **Psychoacoustic model:** Calculate masking thresholds for each band
- 3 **Quantize:** Allocate bits per band — more bits where ear is sensitive
- 4 **Huffman code:** Lossless compression of quantized values

How MP3 Uses Psychoacoustics

The psychoacoustic model drives everything:

- 1 Analyze the audio frame (1152 samples $\approx$ 26 ms)
- 2 Compute the power spectrum via FFT
- 3 Identify tonal (sinusoidal) and noise-like components
- 4 Calculate masking thresholds for each critical band
- 5 Determine the **Signal-to-Mask Ratio (SMR)** per band
- 6 Allocate bits: bands with high SMR get more bits

The Magic

Quantization noise is shaped to stay **below the masking threshold**.

The noise is there, but you **cannot hear it!**

This is called **noise shaping** or **perceptual coding**.

MP3 Quality Comparison

Bitrate	Ratio	Size/min	Quality
320 kbps	4.4:1	2.4 MB	Transparent
192 kbps	7.3:1	1.4 MB	Very good
128 kbps	11:1	960 KB	Good (most common)
64 kbps	22:1	480 KB	Noticeable artifacts

A 50-minute CD album:

- Uncompressed WAV: **530 MB**
- MP3 at 128 kbps: **48 MB** (11× smaller)
- MP3 at 320 kbps: **120 MB** (4.4× smaller)

The MP3 Revolution

Made digital music distribution practical. Napster, iPod, and the entire streaming era followed.

Beyond MP3: Modern Audio Codecs

Codec	Year	Type	Used in
MP3	1993	Lossy	Legacy, portable players
AAC	1997	Lossy	iTunes, YouTube
FLAC	2001	Lossless	Archival, audiophiles
Opus	2012	Both	Discord, WhatsApp, WebRTC

Opus — The Modern Champion:

- Speech and music in one codec
- 6 kbps (speech) to 510 kbps (music)
- Lower latency than MP3
- Open and royalty-free

Takeaway

Opus beats MP3 and AAC at every bitrate. It is the recommended codec for real-time communication and streaming today.

Key Takeaways

- ➊ **Digitization:** Sampling (Nyquist), quantization (bit depth), binary coding
- ➋ **MIDI:** Instructions, not audio — tiny files, editable, synth-dependent
- ➌ **Lossless compression:** Entropy is the limit; Huffman, arithmetic, LZW
- ➍ **Lossy compression:** Rate-distortion tradeoff; quantization is the core
- ➎ **Transforms:** DCT and wavelets concentrate energy for efficient coding
- ➏ **Speech coding:** ADPCM (prediction), vocoders (modelling), CELP (hybrid)
- ➐ **Psychoacoustics:** Masking and hearing thresholds let us discard inaudible data
- ➑ **MP3:** Filter bank + psychoacoustic model + quantization + Huffman = 11:1 compression with near-CD quality

Thank You!

Questions?

“If you can't hear it, don't waste bits on it.”

— The philosophy behind perceptual audio coding